Mallow Technologies

Laravel Developer

Mini Task — Take-Home Assignment

Thank you for taking the time to work on this assignment. Please read the full brief before you start. If anything is ambiguous, use your best judgment, make a reasonable assumption, and document it in your README — we're as interested in how you handle ambiguity as in the result.

Brief: Store Order & Inventory Mini-System

Build a small Laravel application that lets a retail counter record customer orders against a product catalog and keep stock in sync.

Scope

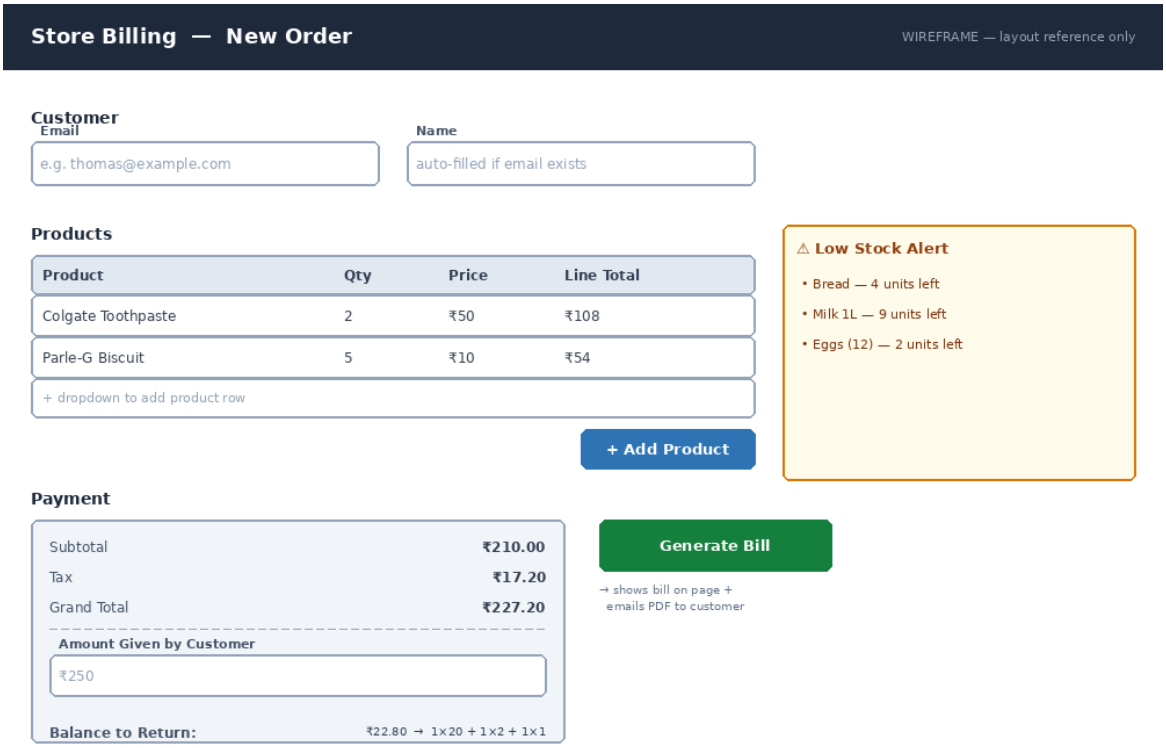
- Products: name, unique code, price per unit, tax percentage, stock on hand.
- Customers: name, email (unique).
- Orders: one customer, one or more product lines (product + quantity), computed subtotal, tax and grand total.
- Seed the database with sample products and customers using factories/seeders.

Functional Requirements

1. Design a normalized schema (migrations) for the above, adding any supporting tables you judge necessary.
2. API endpoint to create an order: accepts customer email/name and a list of {product_id, quantity}; validates stock availability, computes totals including tax, deducts stock, and returns the created order.
3. API endpoint to fetch a customer's order history by email.
4. API endpoint returning products below a configurable low-stock threshold.
5. Dispatch a queued job on order creation that simulates sending an order-confirmation email (a log entry or fake mailer is fine — no real SMTP needed).
6. Write feature/unit tests (PHPUnit or Pest) covering order creation, including at least one edge case (e.g. insufficient stock).
7. Ensure the stock check-and-deduct step is safe under concurrent requests — if two orders for the same product arrive at nearly the same time and only one unit is left, exactly one should succeed and the other should fail cleanly (no overselling).

Suggested UI Reference (Wireframe)

This is a low-fidelity layout reference — it shows the expected screen structure and key fields so you can visualize scope before you start. Visual styling, colors and exact layout are entirely up to you; we're evaluating the data and logic underneath, not pixel-perfect design.



What We're Looking For

- Correct use of Eloquent relationships, migrations and form-request validation.
- Clean, readable code organized the Laravel way (controllers thin, logic in services/actions as appropriate).
- Sensible use of a queued job — shows you understand async processing, not just direct calls.
- Test coverage that reflects how you think about edge cases, not just the happy path.
- A README documenting setup steps and any assumptions made.

AI-assisted development (Copilot, Cursor, Claude, etc.) is welcome and encouraged for this task — we're as interested in how well you use these tools as in the output itself. See the Prompt Log requirement below.

What to Submit

- Codebase: a GitHub repo (private, shared with us) containing the full Laravel application.
- README.md: setup/run instructions, plus a clear write-up of the assumptions and design decisions you made. If any requirement is ambiguous, make a reasonable assumption, proceed, and document it here — don't stop and wait to ask.
- Prompt Log: if you used AI-assisted tools (Copilot/Cursor/Claude etc.), include screenshots of the actual prompts you used (from your chat/IDE panel) — so we can see exactly what was asked. Save these in a /prompts folder or embed them in the README.
- Screen Recording: a short (5–10 min) video walkthrough of the working application, narrated by you — walk us through what you built and why. A silent screen capture isn't sufficient.

Timebox & Submission

- Timebox: 3 days elapsed time from receiving this brief.
- Please share your GitHub repo link, README, prompt screenshots (if applicable) and narrated screen recording with [recruiter/HR email] once ready.
- If anything in this brief is unclear, think it through, make the call you believe is most reasonable, and document your reasoning in the README.md — we're evaluating your judgment under ambiguity, not whether you guessed our exact intent.